AHMED ALVEE

Solutions Architect | AI & Automation Engineering
Rotterdam, Netherlands | +31 6 4309 7073 | ahmedalv94@gmail.com
linkedin.com/in/alveeahmed | github.com/alvee1994 | alvee1994.github.io/portfolio

Summary

Solutions architect with a background across robotics research, C++ medtech software, healthcare data integration on AWS and LLM workflow design. Built AWS data pipelines and LLM workflows, took them live with customers on site, led the first overseas deployment at Mesh Bio and led a team of 3 solutions engineers. Designed an AI monitoring workflow for Truvalu as part of a 5-person MBA team in the RSM consulting project. AWS Certified Solutions Architect – Associate. Completing an MBA at Rotterdam School of Management.

Skills

Cloud: AWS (Lambda, Step Functions, SQS, SNS, EventBridge), Terraform, Docker (deploying containerized applications); AWS architecture skills transferable to Azure

AI and LLM: Generative AI (LLMs, VLMs), AI agents, RAG and vector search (ChromaDB, Cohere embeddings), Model Context Protocol (MCP), Amazon Bedrock, OpenRouter, vLLM, Hugging Face, LangGraph (supervisor agents), LangChain, Gemini, Claude, Pydantic, system prompting, guardrails, model evaluation, n8n (low-code), Claude Code, Cursor

Integration: REST APIs, webhooks, OAuth (API authentication and identity, Microsoft Graph API), Postman, HL7, MLLP, DICOM

Languages and tools: Python, C++, MongoDB, Git, Linux, Qt, ROS

Ways of working: Agile teams, sprints, Scrum, stakeholder management, responsible AI, AI discovery, cross-functional teams

Professional Experience

Mesh Bio Pte Ltd — Singapore, SG

Solutions Architect (prev. Senior Solutions Engineer) — 05/2022 – 12/2025

Singapore health-tech company working with private healthcare providers. Its most widely adopted product was a health screening workflow that scored each patient's risk and built screening reports. Built the pipelines behind it.

- Per-submission time improved from about 1 hour to 6-10 minutes, including human review, for 30+ clinics, by choosing LLMs to turn messy input into schema-aligned output and building the pipeline from scratch (LangGraph, Claude on Bedrock, Pydantic, ChromaDB); a human reviewer checked every file.
- Developed guardrails for the AI pipeline: system prompting, a LangGraph supervisor agent framework and a configurable Pydantic validation layer, with strict submission criteria as a responsible AI control.
- Built prototypes, proofs of concept and MVPs, ran pilots with teams in Indonesia before wide deployment, and extended the pipeline to run on premises (not cloud) as an MVP for a pilot with an Indonesian client.
- Chose a vision language model (VLM) for a medical identification pilot with a major healthcare provider, and deployed, tested and evaluated frontier and open-weight models (OpenRouter, vLLM, Hugging Face, Amazon Bedrock).
- Onboarding time improved from 3-5 days to under 1 day for each new lab and clinic combination, by redesigning a hardcoded Lambda per combination into a config-driven AWS pipeline (Lambda, SNS, SQS, EventBridge Pipes, Step Functions, HL7 integration) that handled about 10,000 records an hour in Singapore and stayed inside Lambda and MongoDB connection limits.

- Turned the company's first overseas customer into a lighthouse account that opened follow-on regional projects, by recommending a headless API-based workflow, cutting scope to meet the deadline, and leading 3 weeks of on-the-ground deployment.
- Joined technical sales calls to map customer tech stacks and integration points, and owned the integration APIs, built ahead of demand with Postman documentation.
- Defined non-functional requirements with technical and non-technical teams (rate limiting, concurrent requests), drew workflow and data-flow diagrams, and designed the API for embedding Mesh Bio's online platform in a CMS used by a third of Singapore's private healthcare providers.
- Promoted to Solutions Architect; hired (2024 to mid 2025), coached, trained and supervised a team of 3 solutions engineers as they took over and expanded the APIs and pipelines.

Creative Medtech Solutions Pte Ltd — Singapore, SG
Software Engineer — 09/2021 – 04/2022
Singapore medtech company building a robotic system that ablates cancer cells with high intensity focused ultrasound (HIFU).

- Owned the annotation and user-interaction layer of the imaging system for a HIFU cancer-ablation robot: marking and measuring ablation targets, and DICOM export.

Singapore University of Technology and Design — Singapore, SG
Research Assistant — 02/2018 – 08/2021
University research lab building robotics and wearable systems, working with teams of research assistants and PhD students.

- Mapped 2 floors and 5 rooms in under 6 minutes with a 12+ robot system, owning its locomotion code.
- Recognized 7 hand gestures to command unmanned vehicles with an ML pipeline (Python, ROS, StretchSense).
- Contributed C++ and hardware engineering to 2 more robotics builds: a legged mapping robot and a custom PCB for 3D-knitted wearables.

MBA Consulting Project

RSM Living Management Project — Rotterdam, Netherlands
AI Strategy Consultant (Client: Truvalu) — 06/2026 – 07/2026
MBA consulting project for Truvalu, a Netherlands-based impact investor supporting agri-food SMEs in emerging markets.

- Defined the problem with the client (AI discovery), then designed a 5-stage AI monitoring workflow with a 5-person MBA team, projected to improve monitoring time for an impact investor by about 45%.
- Engineered and demoed AI tools and agents (Gemini, Claude, Tally.so) as proofs of concept and an MVP spanning all 5 workflow stages, from data clean-up to reporting.
- Co-led the final client presentation to Truvalu leadership.

Education

Rotterdam School of Management, Erasmus University — Rotterdam, Netherlands
Master of Business Administration (MBA) — 01/2026 – 03/2027

Singapore University of Technology and Design — Singapore, SG
BEng (Honours), Engineering Product Development — 05/2014 – 08/2017

Certifications

- AWS Certified Solutions Architect – Associate, Amazon Web Services (10/2023).